The subcaptionlite package

Subfigures and subcaptions without the caption package

Munehiro Yamamoto
https://github.com/munepi/subcaptionlite

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1 Introduction

The subcaption package needs the caption package, which redefines `\@makecaption` of the class. Thus it cannot be used with a class that has its own design of captions: loading it changes the captions of the whole document.

The subcaptionlite package provides the `subfigure` environment, `\subcaption` and `\subcaptionbox` with the same syntax as subcaption, without touching `\@makecaption` of the class. Figure 1 is made with it.

Supported formats/engines pdfLaTeX, XeLaTeX, LuaLaTeX (with or without LuaTeX-ja), pLaTeX and upLaTeX (with dvipdfmx). The package requires LaTeX2e 2020-10-01 or later and loads etoolbox.

2 Usage

```
\usepackage{subcaptionlite}

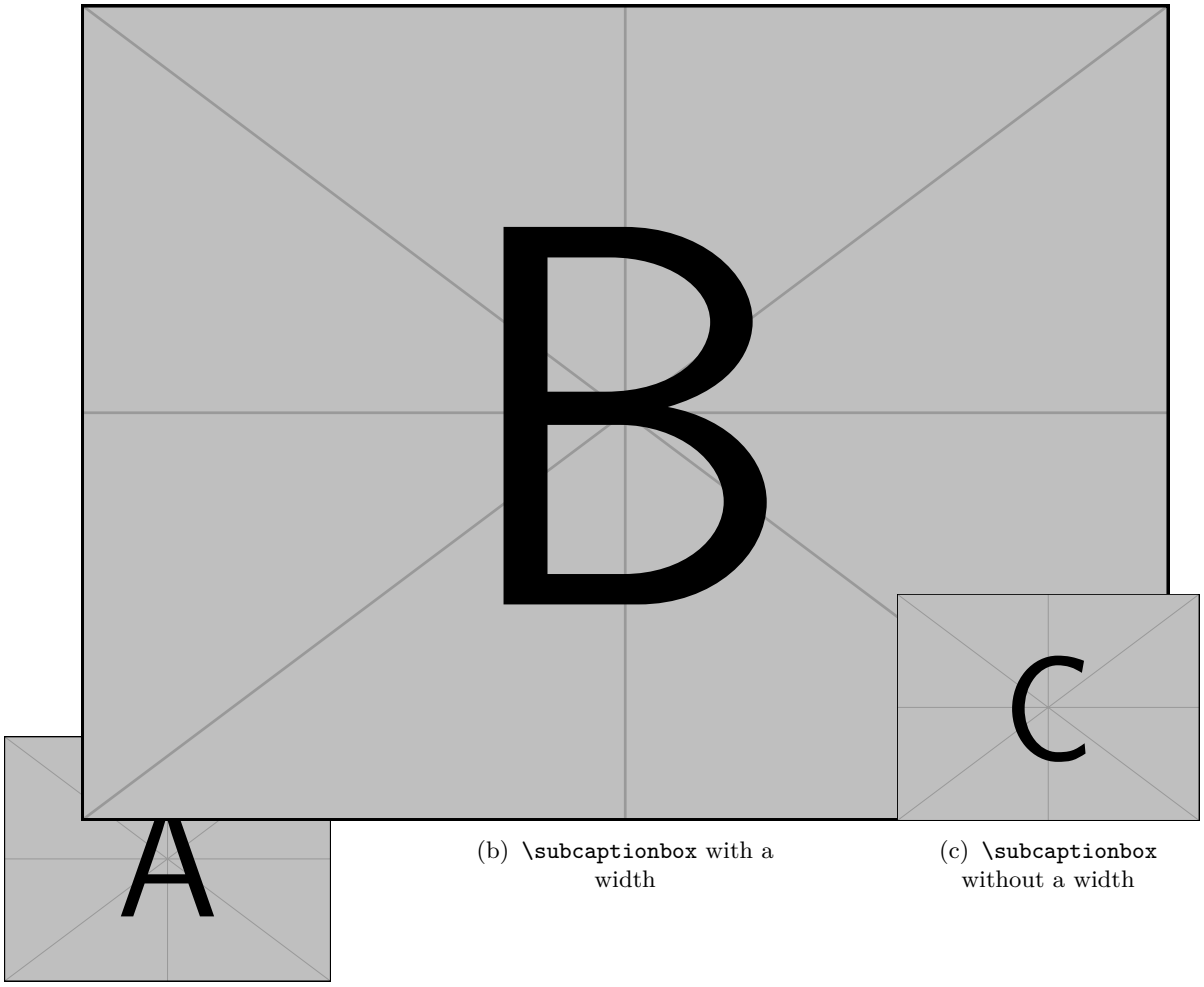
\begin{figure}
  \centering
  \begin{subfigure}{.3\linewidth}
    \centering\includegraphics[width=.9\linewidth]{example-image-a}
    \caption{The subfigure environment}\label{fig:example-a}
  \end{subfigure}\hfill
  \subcaptionbox{\cs{subcaptionbox} with a width\label{fig:example-b}}%
    [.3\linewidth]{\includegraphics[width=.9\linewidth]{example-image-b}}\hfill
  \subcaptionbox{\cs{subcaptionbox} without a width\label{fig:example-c}}{%
    \includegraphics[width=.25\linewidth]{example-image-c}}
  \caption{Three subfigures}\label{fig:example}
\end{figure}
```

`\begin{subfigure}[`⟨*pos*⟩**`]{`**⟨*width*⟩**`}`** A `minipage` of ⟨*width*⟩ (⟨*pos*⟩: `c` by default), inside which `\caption` and `\caption*` mean `\subcaption` and `\subcaption*`.

`\subcaption{`⟨*text*⟩**`}, \subcaption*{`**⟨*text*⟩**`}`** A numbered or unnumbered subcaption, e.g. at the bottom of a `minipage`. It is centered and set in `\small`.

`\subcaptionbox{`⟨*caption*⟩**`}[`**⟨*width*⟩**`]{`**⟨*contents*⟩**`}`** Puts ⟨*contents*⟩ centered in a `minipage` of ⟨*width*⟩, followed by `\subcaption{`⟨*caption*⟩`}`. Without ⟨*width*⟩, the natural width of ⟨*contents*⟩ is used. If ⟨*contents*⟩ is wider than ⟨*width*⟩, the `minipage` keeps ⟨*width*⟩. Boxes are aligned at the bottom.

`\captionsetup[`⟨*type*⟩**`]{`**⟨*options*⟩**`}`** Accepted and ignored, so that the settings for the caption package can be left in the document.



(a) The subfigure environment

(b) `\subcaptionbox` with a width

(c) `\subcaptionbox` without a width

Figure 1: Three subfigures

Numbering and references The counter `subfigure` is reset by `figure`. `\thesubfigure` is `(\alph{subfigure})` and `\p@subfigure` is `\thefigure`, so that `\ref` gives "1(a)" and `\Cref` of cleveref gives "Figure 1(b)". With hyperref, each subfigure has its own anchor (`\theHsubfigure`).

The `\caption` of the figure may come after the subcaptions (as in Fig. 1) or before them. When a subcaption comes first, it steps the figure counter in advance and the `\caption` that follows does not step it again, as the subcaption package does; without this, the subfigures would get the number of the previous figure. A figure with subcaptions and no `\caption` is numbered as well.

3 Customization

The following macros can be redefined in the class or in a local style (`\makeatletter` is needed in the preamble of a document).

Macro	Default
`\subcaption@font`	`\small`
`\subcaption@sep`	the space between the label and the text: `\hspace{.5zw}` with pTEX, `.5\zw` with LuaTEX-ja (or a class that defines `\zw`), `.5em` otherwise
`\subcaption@skip`	the space between the subfigure and its subcaption: `\vskip.25\baselineskip`

The engine is checked when the subcaption is typeset, so LuaTEX-ja may be loaded after subcaptionlite. The label itself is `\thesubfigure`. For example,

```
\makeatletter
\renewcommand*\subcaption@font{\footnotesize\sffamily}
\renewcommand*\subcaption@sep{\enspace}
\makeatother
\renewcommand\thesubfigure{\alph{subfigure})}
```

4 Limitations

- Subcaptions are for figures only: there is no subtable.
- subcaptionlite cannot be used together with the caption or subcaption package.

5 Tests

`make test` compiles the test files twice with pdfLATEX, XeLATEX, LuaLATEX, pLATEX and upLATEX (+ dvipdfmx) and checks the log. The helpers in `test-common.tex` check the label that `\label` would write, the references, the counters, the widths of the boxes, the anchors of hyperref, and that `\@makecaption` of the class is unchanged; the log must have `SCL-RESULT OK` and no error or warning.

Test	What it covers
`test-basic.tex`	`subfigure`, `\subcaption`, `\subcaptionbox`, the caption of the figure below/above the subfigures or none, `figure*`, `\ref`, the label separator
`test-hyperref.tex`	`test-basic.tex` with hyperref (loaded after and before the package) and cleveref
`test-japanese.tex`	Japanese classes (ltjsarticle, jsarticle, ujarticle), LuaTEX-ja loaded after the package

6 History

v1.0 (2026/10/26) First public release. Support every engine; fix the figure number when the caption of the figure comes before the subcaptions, and after a figure whose subfigures follow its caption.

7 License

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